Cryptanalysis of Shift Cipher

Brute Force, Dictionary Scoring and Chi-Square Analysis

Course: Cryptography Laboratory (22CPP307)
Instructor: Dr. Meenakshi Tripathi
Group: 9

Team Members:

Amit Kumar (2024UCP1562)
Guttula Pooja (2024UCP1717)

1. Algorithms of Cryptanalysis

1.1 Shift Cipher (Caesar Cipher)

The Shift Cipher encrypts each letter of the plaintext by shifting it forward by a fixed key k positions in the alphabet. The encryption and decryption functions are:

E(x) = (x + k) mod 26
D(y) = (y - k) mod 26

Non-alphabetic characters (spaces, punctuation) are preserved unchanged. The key space is limited to 26 possible values (0-25), making the cipher vulnerable to brute-force attacks.

1.2 Chi-Square Statistical Analysis

The chi-square attack exploits the known frequency distribution of letters in the English language. The algorithm proceeds as follows:

- For each candidate key k (0 to 25), decrypt the ciphertext using key k
- Count the frequency of each letter (A-Z) in the decrypted text
- Compute the chi-square statistic against expected English letter frequencies
- Select the key that produces the lowest chi-square score (best fit to English)

The chi-square formula is:

X^2 = SUM[(observed_i - expected_i)^2 / expected_i]

where observed_i is the actual count of letter i in the decrypted text and expected_i is the expected count based on standard English letter frequencies (e.g., E = 12.70%, T = 9.06%, A = 8.17%).

1.3 Brute Force with Dictionary Scoring

The dictionary attack uses brute force combined with a word-matching heuristic. The algorithm proceeds as follows:

- Load a dictionary of common English words from a text file
- For each candidate key k (0 to 25), decrypt the ciphertext using key k
- Split the decrypted text into words and count how many match dictionary entries
- Select the key that produces the highest dictionary match score

Both algorithms iterate through all 26 possible keys, making them O(26 x n) where n is the length of the ciphertext. This is computationally trivial.

2. Results Table

#	Plaintext	Actual Key	Chi-Sq Key	Dict Key	Chi-Sq Correct?	Dict Correct?
1	HELLO WORLD	3	6	3	No	Yes
2	ATTACK AT DAWN	5	5	5	Yes	Yes
3	CRYPTOGRAPHY IS FUN	10	23	10	No	Yes
4	THIS IS A SIMPLE TEST	7	7	7	Yes	Yes
5	THE QUICK BROWN FOX...	12	12	12	Yes	Yes
6	MEET ME AT THE STATION	4	4	4	Yes	Yes
7	COMPUTER SECURITY...	8	8	8	Yes	Yes
8	INFORMATION SECURITY	15	15	15	Yes	Yes
9	SHIFT CIPHER IS EASY...	6	6	6	Yes	Yes
10	THIS IS A CRYPTO LAB	20	20	20	Yes	Yes

Chi-Square Accuracy: 8/10 (80%) **Dictionary Accuracy: 10/10 (100%)**

3. Comparison of Attack Methods

The two attack methods were compared across all 10 testcases:

Criterion	Chi-Square Analysis	Dictionary Scoring
Accuracy	8/10 (80%)	10/10 (100%)
Short text reliability	Poor (fails < 20 chars)	Excellent
Long text reliability	Excellent	Good
External data needed	Letter frequency table	Dictionary word list
Computational cost	O(26 x n)	O(26 x n x w)
Statistical basis	Frequency distribution	Word recognition

The dictionary attack outperformed chi-square in overall accuracy, particularly on short texts. However, chi-square does not require an external word list and works purely on statistical properties, making it more language-agnostic in theory. For practical use, combining both methods would yield the most reliable results.

4. Failure Analysis

Testcase 1: HELLO WORLD (key=3)

Chi-square predicted key 6 instead of 3. The plaintext contains only 10 letters. With such a small sample, the observed letter frequencies deviate significantly from the expected English distribution for all candidate keys. The chi-square statistic cannot reliably distinguish the correct key from competing candidates when the sample size is this small.

Testcase 3: CRYPTOGRAPHY IS FUN (key=10)

Chi-square predicted key 23 instead of 10. This text has 16 letters, still below the threshold where chi-square analysis becomes reliable. Additionally, the word 'CRYPTOGRAPHY' has an unusual letter distribution (no E, the most common English letter), which further confuses the statistical method.

Suggested Improvements

- Combine chi-square with dictionary scoring as a weighted ensemble approach
- Use bigram/trigram frequency analysis alongside single-letter frequencies
- Apply index of coincidence as a secondary validation metric
- Use a larger, more comprehensive dictionary file for better word coverage

5. Observations

- The dictionary attack achieved 100% accuracy across all 10 testcases, proving its reliability for texts containing common English words.
- The chi-square attack performed well on texts with 20+ characters but failed on shorter texts where letter frequencies do not converge to the expected English distribution.
- The pangram testcase (THE QUICK BROWN FOX...) had a high chi-square score (109.3) even when correctly identified, because pangrams have a more uniform letter distribution than typical English.
- Both attacks complete in constant time relative to key space (26 iterations), making the Shift Cipher trivially breakable regardless of the attack method used.
- The two methods are complementary: chi-square excels at statistical pattern matching on longer texts, while dictionary scoring excels at semantic word recognition even on short texts.

6. Conclusion

This assignment demonstrated the practical cryptanalysis of the Shift Cipher using chi-square statistical analysis and brute-force dictionary scoring. The key findings are:

- The Shift Cipher is inherently insecure because its key space of only 26 values makes exhaustive search trivial for any automated tool.
- Statistical methods like chi-square are powerful for longer ciphertexts but unreliable for short texts (fewer than approximately 20 characters).
- Dictionary-based scoring provides a robust complement that works effectively even on very short texts, as long as the plaintext contains recognizable English words.
- A combined approach using multiple cryptanalysis techniques would produce the most reliable results in practice.
- Modern encryption algorithms must use significantly larger key spaces and more complex mathematical transformations to resist automated cryptanalysis.